The Venn diagram

Which features two or three samples, or two or three conditions, have in common — and which are unique to one of them.

AT A GLANCE

Where	Comparison → Venn Diagram
Data levels	Protein, phosphosite, phosphoprotein
Needs	An uploaded dataset and its annotation, and two different samples or conditions; a third is optional.
Data source	Annotated, optionally normalised data. Comparison has no Filtered or Imputed choice — an imputed matrix would describe filled values rather than observed detection.
Membership	Quantified, not regulated: a feature is in a set when it has a value there.
Sets	Two or three, of samples (Single) or conditions (Condition).
Table	Yes, below the plot: every region and every total, downloadable as CSV.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

1 What the plot answers

Whether two runs, or two conditions, saw the same features. Nothing more than that: the diagram is built from presence and absence, so a feature sits in the intersection if both selections measured it, however differently. It is a statement about coverage, and it answers the question a volcano plot cannot — what was never in the comparison at all.

In Condition mode a feature belongs to a condition when at least one of its replicates measured it. The sets are therefore unions, and a condition with six replicates has more chance of including a marginal feature than one with three; part of any unique wedge is that difference rather than biology.

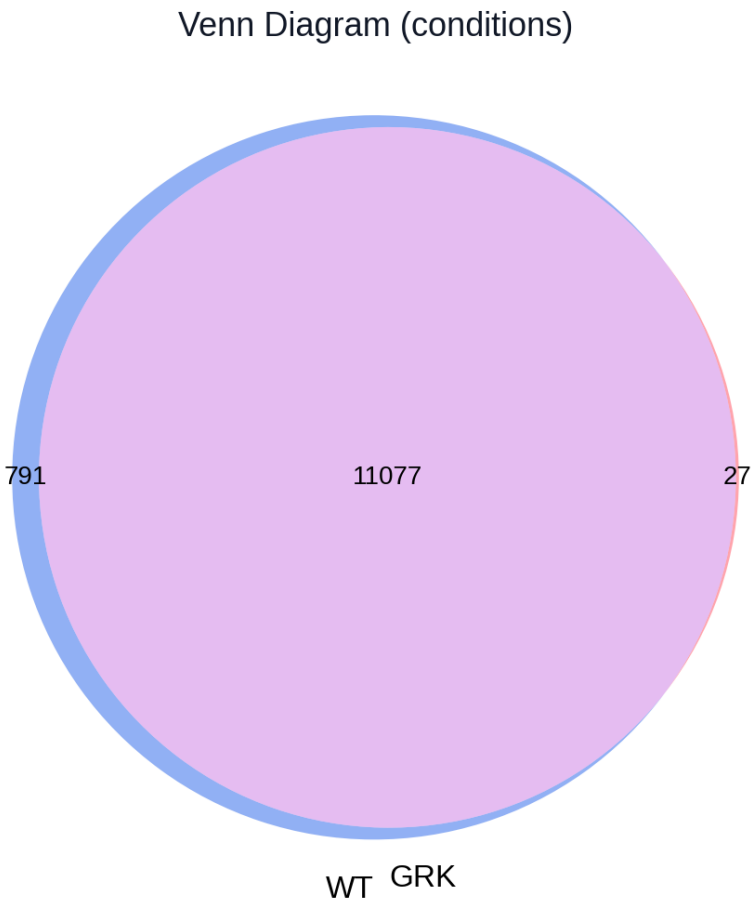


Figure 1.1. Mode Condition, phosphosite level on the example project: the sites quantified in at least one of the five WT samples against those quantified in at least one of the five GRK samples. 11,077 are in both, 791 only in WT and 27 only in GRK. The circles are drawn to scale, so near-total overlap looks like one circle inside another — which is itself the finding. Read the counts rather than the areas.

> **NOTE**
>
> This figure is drawn at phosphosite level on purpose. At protein level the same comparison gives 7,457 shared proteins against 3 and 30 unique ones: the two circles coincide exactly, the wedges are too thin to see and the two set names print on top of each other. A Venn is worth drawing when the sets differ; when they do not, the count of shared features is the whole answer and the table below says it more clearly.

Reading it

- A region: the number of features quantified in exactly that combination of selections — the centre of a three-set figure is the features all three have.
- A set name under each circle: the sample or condition name, or the alias if one was typed.
- A wedge with no overlap: features only that selection measured.
- The title says which mode produced the figure, Venn Diagram (samples) or Venn Diagram (conditions), because the same three names can mean either.

> **EASY TO GET WRONG**
>
> A set is a set of *identifiers*, not of rows. Two rows carrying the same protein name or the same collapse key are one element of the set, so the total under a selection can be smaller than the number of rows it has values in. Missing identifiers are excluded; literal blank strings or malformed identifiers need cleaning rather than being treated as biological features. The Tables page counts rows; this page counts identifiers, and the two are equal only when the identifier column is unique.

2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Which dataset to compare: Protein, Phospho or Phosphoprotein. Only the levels actually uploaded are offered.	Protein	page
Mode	Single makes one set per sample. Condition makes one set per condition, pooling its replicates.	Single	page
Sample 1 / Condition 1	The first set. The label follows **Mode**.	the first sample or condition of the annotation	page
Sample 2 / Condition 2	The second set. It may not repeat the first.	the first one that differs	page
Sample 3 (optional) / Condition 3 (optional)	The third set. Left at None the figure has two circles; chosen, it has three and must differ from both others.	None	page
Alias 1, Alias 2, Alias 3 (optional)	Replace the set names under the circles and in the table. The sets themselves are unaffected.	empty	page

Controls on this page (continued)

Control	What it does	Default	Where
Width, Height, DPI	The size of the *downloaded* figure. The browser scales the rendered figure to fit the page. Width and height are held between 1 and 60 cm, the resolution between 72 and 600.	15 cm × 12 cm, 300	page
Download format	PNG, JPG, WEBP, SVG or PDF.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Venn sets	One colour per set, applied at half opacity so that the overlaps darken. The figure uses blue #1f77b4, orange #ff7f0e and green #2ca02c until a swatch is changed.	as drawn	appearance
Title	Shown or hidden, and its text, size, weight, colour and position.	shown	appearance
X axis, Y axis	A Venn diagram has no axes and carries no axis text; anything typed here is drawn beside the circles.	empty	appearance
Frame and grid	Which of the four spines are drawn, and the grid behind the circles. This figure opens with neither.	no spines, no grid	appearance
Font	One face for the whole figure, or one per text.	plot default	appearance
Background	Theme, paper colour and transparency.	Default, white, opaque	appearance

WORTH KNOWING

The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here. The per-set colour pickers the earlier manual listed beside the plot are the **Venn sets** card here.

3 How it is computed

HOW IT IS COMPUTED

For each selection, a set of identifiers is built from the loaded matrix. In Single mode, a feature is in the set of sample j when x_{ij} is a number and not missing:

$$S_j = \{\, \mathrm{id}_i \;:\; x_{ij} \neq \mathrm{NA} \,\}.$$

In Condition mode the test is taken over the condition's replicates and satisfied by any one of them:

$$S_c = \{\, \mathrm{id}_i \;:\; \exists\, j \in c,\ x_{ij} \neq \mathrm{NA} \,\}.$$

The identifier is the dataset's own: the protein-name column at protein level, the collapse key or the phosphoprotein name at the phospho levels. The regions are then the ordinary set arithmetic — $S_1 \setminus S_2$, $S_1 \cap S_2$, and for three sets $S_1 \cap S_2 \setminus S_3$ and the rest — and each region's count is printed in it.

ON THE GEOMETRY

The circles are drawn area-weighted: their sizes and, as far as the geometry allows, their overlaps follow the counts. For two sets this is exact; for three it cannot always be satisfied at once and the drawing is an approximation. Measure nothing off the figure — the numbers in the regions and in the table are the exact values, and a very small region may be drawn too small to hold its own label.

Edge cases

- Leaving the third selection empty is the normal two-set case, not an error. Repeating a selection is an error, and the page says so rather than drawing a figure in which one circle contains another.
- A feature missing everywhere is in no set and appears in no region; the totals under the set names therefore do not sum to the number of rows in the dataset.
- Two selections that measured exactly the same features give two concentric circles with an empty outer region, which is what identical coverage looks like here.
- A zero in a linear intensity became a missing value when the data were loaded, so a feature reported as zero counts as absent — the same convention the completeness pages use.

Interpreting overlap

Comparable replicates often share many features, but there is no universal acceptable unique fraction. A large unique region is a reason to inspect coverage, missingness and sample identity, not proof of a failed run. Between conditions, a feature just above a detection threshold

on one side and below it on the other can appear unique without evidence of regulation. Use single-feature views to inspect its measurements. A volcano can test features with enough values in both groups; an entirely one-sided feature does not become statistically significant merely because it occupies a unique Venn region.

4 The table below the plot

One row per region, then one row per set total. With three sets there are seven regions and three totals; with two, three regions and two totals.

Columns in this table

Column	What it holds
Region	The region, written from the set names or aliases: A only, A & B, A & B & C, and Total A for the whole of a set.
Count	How many identifiers fall in that region. The Total rows are the size of each set, which is the sum of every region that set takes part in.

5 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

PYTHON

```
data_by_protein = data.set_index("ProteinNames", drop=False)
fig = overlap_plot_condition(data_by_protein, meta, "WT", "GRK", data_scale="log2")
```

R

```
data_by_protein <- data
rownames(data_by_protein) <- data_by_protein$ProteinNames
fig <- overlap_plot_condition(data_by_protein, meta, "WT", "GRK", data_scale = "log2")
```

The indexing line is not decoration. The sets are built from the frame's *index*, so a frame passed in as it was read gives sets of row numbers, every one of them distinct, and a diagram whose regions are counts of rows rather than of identifiers. Setting the index to the identifier

column first — ProteinNames at protein level, PTM_Collapse_key and Phosphoprotein at the two phospho levels — is what makes the notebook figure the one the page draws. overlap_plot_single is the Single mode counterpart and takes sample column names in place of the annotation and the two conditions; in either, a third selection is the optional argument that follows the second. The notebook titles the figure Overlap rather than naming the mode.

6 Exporting

The download reproduces the figure at the width, height and resolution set on the page, which is why those three controls do not change what is on screen. A Venn diagram is line art with a few labels, so PDF is the sensible choice wherever it can be used: the circles stay sharp at any size. The table underneath downloads separately as CSV, and it is the only complete record of the counts — a region too small to carry its label on the figure still has its row there.

Behind this page

Function	What it produces
overlap_plot_single	The two- or three-set diagram for individual samples
overlap_plot_condition	The same with the replicates of each condition pooled
venn_png	The figure behind the page, with the set colours and the mode in its title
venn_table	The region and total counts under it